# N-of-1 Trials and Precision Medicine: A Detailed Lexicon

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A working glossary of the terminology used across the N-of-1 / single-case trial and precision-medicine literature. Each entry pairs a technical term with a plain-language definition pitched at an undergraduate statistics major, and lists the alternative names the same concept carries in different fields (biostatistics, psychometrics / single-case research, and precision / personalized medicine).

Provenance. Terms were mined from the full text of N-of-1 and precision-medicine articles in the project Zotero library (`nof1` collection, 72 curated articles; full text obtained and machine-read for 42). The statistical and psychometric supplement was compiled from standard methodology references. See the closing Provenance and epistemic status section for scope and caveats.

How to read the synonym column. Where biostatistics, psychometrics, and precision medicine use different names for the same underlying concept, all variants are listed together so the same idea can be recognised whatever vocabulary a given paper adopts. A condensed same-concept / different-name map appears in Section 12.

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## 1. Trial designs and architectures

| Term | Plain-language definition | Synonyms / aliases (by field) |
|---|---|---|
| N-of-1 trial | A controlled experiment run inside a single patient, who is switched between treatments multiple times so the best treatment for that one person can be estimated. | single-patient trial; single-subject trial; single-case experimental design (SCED, psychology); individualized / idiographic clinical trial; one-person trial; n=1 RCT |
| Aggregated N-of-1 trial | A set of separate single-patient trials pooled together to estimate both a population-average effect and sharpened individual effects. | series / combined / pooled N-of-1 trials; aggregated personalized studies; multiple N-of-1 study |
| Single-case experimental design (SCED) | The behavioural-science family of designs that study one subject repeatedly across baseline and treatment phases; N-of-1 trials are the randomized-crossover member of this family. | single-subject design; single-case design; single-subject research |
| Crossover trial | A trial in which each participant receives the treatments in sequence over separate periods, serving as their own control. | cross-over / change-over trial; within-subject / within-patient design; repeated-measures comparison |
| AB/BA (2x2) crossover | The simplest crossover: half the subjects get A then B, half get B then A, in two periods. | two-period two-treatment design; classical crossover |

| Term | Plain-language definition | Synonyms / aliases (by field) |
|---|---|---|
| Multiple-crossover design | A crossover in which the patient switches back and forth between treatments several times rather than once. | multi-crossover; repeated-period crossover; double crossover |
| Parallel-group trial | A trial in which each participant receives only one treatment and group averages are compared between people. | parallel-group RCT (PG-RCT); between-participant / between-patient design |
| Randomized controlled trial (RCT) | A study that assigns participants to treatments by chance to give an unbiased comparison. | controlled trial; randomized trial |
| AB / ABAB design | Single-case notation for alternating a baseline/treatment (A) and an alternate treatment (B); ABAB introduces, withdraws, and reintroduces treatment. | reversal design; withdrawal design; challenge-withdrawal design |
| Multiple-baseline design | A single-case design that staggers when treatment starts across people, settings, or behaviours to rule out coincidental change. | staggered-baseline design |
| Changing-criterion design | A single-case design in which the success threshold is shifted in steps and the outcome is shown to track each new target. | – |
| Alternating-treatments design | A single-case design that rapidly switches among two or more treatments within one subject to compare them. | multi-element design |
| Open-label (OL) design | A study or phase in which both patient and staff know which (active) treatment is given. | unblinded design |
| Blinded discontinuation (BDC) | A phase in which active treatment is secretly replaced by placebo to see whether benefit fades. | open-label + blinded discontinuation (OL+BDC) |
| Micro-randomized trial (MRT) | A within-person trial (often via phone or wearable) that randomizes a brief intervention many times to build just-in-time adaptive interventions. | micro-randomised trial |
| Interrupted time-series design | Repeated measurements on one unit compared in level/slope before and after a defined intervention point. | ITS; single-case time series |

| Term | Plain-language definition | Synonyms / aliases (by field) |
|---|---|---|
| Adaptive design | A trial allowing prospectively planned changes (e.g., dropping arms, reallocation) based on accumulating data. | response-adaptive / sequential design; group-sequential design |
| Play-the-winner / response-adaptive randomization | An adaptive scheme that shifts the randomization odds toward the treatment performing better. | response-adaptive randomization |
| Biomarker-stratified trial | A trial that splits randomization by a baseline biomarker and tests the treatment within each biomarker group and overall. | biomarker-driven / stratified design |
| Enrichment / adaptive-enrichment design | A trial that preferentially enrols (or, after an interim look, narrows to) the patients most likely to respond. | population enrichment; adaptive enrichment |
| Basket trial | One treatment tested across multiple diseases that share a common molecular target. | – |
| Umbrella trial | Multiple treatments matched to different biomarkers within a single disease. | – |
| Platform trial | A standing trial framework evaluating multiple treatments against a shared control, adding or dropping arms over time. | – |
| Two-stage / multi-stage design | A trial run in phases with an interim analysis between them that can change later conduct. | two-stage adaptive design |

## 2. Periods, sequence, carryover, and washout

| Term | Plain-language definition | Synonyms / aliases (by field) |
|---|---|---|
| Treatment period | A block of time during which a patient stays on one assigned treatment before switching. | period; treatment block; phase; intervention administration period |
| Cycle | One full pass through the treatments being compared (e.g., one A-then-B pair). | pair; treatment pair; block |
| Sequence | The full ordered list of treatment periods a participant goes through. | treatment sequence; allocation sequence |

| Term | Plain-language definition | Synonyms / aliases (by field) |
|---|---|---|
| Washout period | A treatment-free gap inserted so leftover effects of the prior treatment disappear before the next is assessed. | wash-out; clearance / rest interval |
| Active (analytical) washout | Switching directly to the next treatment but not measuring until the previous drug's effect has gone, used when stopping treatment would harm the patient. | analytical washout |
| Run-in / titration phase | A preliminary phase before randomization used to stabilize dose, confirm eligibility, clear prior drugs, or assess adherence. | lead-in; stabilization phase; dose titration / escalation |
| Baseline phase | The pre-treatment stretch of repeated measurements used as the comparison reference. | pre-intervention phase; phase A |
| Carryover effect | When a treatment's effect persists into and contaminates the following period. | carry-over; residual / lingering effect |
| Simple carryover | A carryover whose size is the same regardless of which treatment follows; assumed to last only one period. | first-order carry-over |
| Complex carryover | A carryover whose size or direction depends on the specific order or pairing of treatments. | treatment-sequence-dependent carryover |
| Behavioural / psychometric carryover | A non-pharmacological carryover where earlier treatment changes habits, adherence, or how a patient remembers rating themselves, mimicking a drug carryover. | behavioural carry-over; recall carryover |
| Period effect | A systematic change in outcome tied to which time period it is, independent of treatment. | time / secular / nominal trend; cycle effect |
| Order / sequence effect | A bias arising because the order in which treatments are given affects the response. | treatment-ordering effect |
| Treatment-by-period interaction | When the treatment difference is not the same across periods, often a sign of carryover or instability. | period-by-treatment interaction |
| Counterbalancing | Building a balanced (non-random) treatment order such as ABBA to offset known time trends. | counterbalanced design |

| Term | Plain-language definition | Synonyms / aliases (by field) |
|---|---|---|
| Condition- vs time-dependent crossover rule | Whether to switch treatments once the patient returns to baseline (condition-dependent) or after a fixed wait (time-dependent). | crossover rule |
| On/off kinetics | How quickly a drug starts working and wears off, which sets how long a washout must be. | rapid onset / offset; drug half-life |

## 3. Randomization, blinding, and conduct

| Term | Plain-language definition | Synonyms / aliases (by field) |
|---|---|---|
| Randomization | Assigning treatment or treatment order by chance to remove systematic bias. | random allocation / assignment |
| Block (permuted-block) randomization | Randomizing within small balanced groups so each treatment appears equally often and bad runs are avoided. | blocking; restricted randomization; counterbalancing |
| Stratified randomization | Randomizing separately within subgroups (site, sex) to keep those factors balanced. | stratification; minimization; covariate-adaptive allocation |
| Allocation concealment | Hiding the upcoming assignment from those enrolling participants so it cannot be foreseen or manipulated. | concealed allocation |
| Blinding | Concealing which treatment is given from patients and/or assessors to prevent expectation bias. | masking; single / double / triple blind |
| Placebo / placebo effect | An inert comparator made to look identical to the active treatment; the placebo effect is improvement wrongly credited to that inert treatment. | sham / dummy treatment; expectancy response |
| Active comparator / control | The reference treatment (placebo, usual care, or an established drug) against which the test treatment is judged. | comparator; control condition; reference arm |
| Equipoise | Genuine uncertainty about which compared treatment is better, which ethically justifies randomizing. | clinical equipoise |

## 4. Analysis models

| Term | Plain-language definition | Synonyms / aliases (by field) |
|---|---|---|
| Linear mixed-effects model | A regression combining population-level fixed effects with subject-specific random effects, suited to repeated measurements per person. | mixed / mixed-effects model; multilevel model (MLM); hierarchical linear model (HLM); random-effects model; LME; MMRM (mixed model for repeated measures) |
| Multilevel / hierarchical model | A model with parameters nested at several levels (measurements within patients within studies) that share information across levels. | nested model; HLM; intensive hierarchical model |
| Fixed effect | A model term whose coefficient is a single constant assumed to apply to everyone. | population-average / marginal effect |
| Random effect | A model term whose value varies across subjects, treated as a draw from a distribution. | subject-specific effect; random coefficient |
| Random intercept | A per-subject baseline offset letting each individual start at their own level. | subject-specific intercept |
| Random slope | A per-subject deviation in how strongly the outcome changes with a predictor (e.g., over time). | subject-specific slope; random trend |
| Design matrix | The table of predictor values mapping coefficients to observations; separate ones for the fixed (X) and random (Z) parts. | model matrix; X / Z matrix |
| Marginal model | A model describing the population-average outcome rather than a specific individual's. | population-average model |
| Conditional model | A model describing the outcome for a specific individual given their own random effects. | subject-specific model |
| Generalized estimating equations (GEE) | A method for correlated/repeated outcomes that estimates the population-average effect using a chosen working correlation, staying valid even if that correlation is mis-specified. | GEE; Zeger-Liang / marginal approach; PGEE (penalized) |
| Working correlation structure | The assumed (possibly approximate) within-subject correlation pattern GEE uses. | independence / exchangeable / autoregressive working structure |

| Term | Plain-language definition | Synonyms / aliases (by field) |
|---|---|---|
| Generalized linear (mixed) model | A regression for non-normal outcomes (counts, proportions, binary), optionally with random effects. | GLM; GLMM |
| Ordinary least squares (OLS) | Fitting a regression by minimizing the sum of squared residuals. | least-squares regression |
| Generalized least squares (GLS) | Regression that accounts for correlated or unequal-variance errors, giving the best linear unbiased estimate. | weighted least squares (related); BLUE estimation |
| Analysis of covariance (ANCOVA) | A regression that adjusts the outcome for a baseline covariate to improve precision. | baseline adjustment |
| ANHECOVA | Covariate adjustment that includes all treatment-by-covariate interactions, guaranteeing an efficiency gain. | analysis of heterogeneous covariance |
| Change score | The outcome minus its baseline value, a simple (often inefficient) way to adjust for baseline. | difference score; baseline subtraction |
| Proportional hazards (Cox) model | A survival regression in which covariates multiply a baseline rate of an event over time. | Cox regression; coxph |
| Proportional odds model | An ordinal-outcome regression assuming covariate effects are constant across category thresholds. | cumulative logit model |
| Dynamic regression model | A regression that explains the present outcome using lagged (past) values of itself and of predictors, absorbing autocorrelation. | autoregressive distributed-lag model; dynamic modelling |
| Paired t-test | A test comparing two related measurements (e.g., one patient on A vs B) by analyzing their differences. | matched-pairs / Student’s t-test |
| Serial t-test | A t-test for one person’s repeated data that corrects the variance and degrees of freedom for serial correlation. | – |
| Randomization (permutation) test | A significance test that compares the observed effect with effects from many random reassignments of the treatment schedule. | permutation test; randomization-test reference set |

| Term | Plain-language definition | Synonyms / aliases (by field) |
|---|---|---|
| Visual analysis | Judging a single-case treatment effect by inspecting the graphed time series (level, trend, variability) rather than by formal statistics. | visual inspection; graphical analysis |
| Nonoverlap index | A single-case effect size based on how little the treatment-phase data overlap the baseline data. | overlap statistic (see Tau-U, NAP, PND in Section 9) |
| Meta-analysis | Statistically pooling results across studies (or N-of-1 trials) into one overall estimate. | research / quantitative synthesis; mega-analysis |
| Individual participant data (IPD) meta-analysis | Pooling the raw per-patient data across studies, usually via mixed models, rather than just their summaries. | one-stage meta-analysis |
| Network meta-analysis | Combining trials of different treatment pairs to compare treatments never tested head-to-head. | mixed-treatment / indirect comparison |
| Intention-to-treat (ITT) | Analyzing participants by their assigned treatment regardless of adherence, preserving randomization. | ITT (modified ITT excludes those with no outcome data) |
| Per-protocol analysis | Analyzing only participants who followed the protocol, which can introduce bias. | completers / per-protocol population |

## 5. Serial correlation and time series

| Term | Plain-language definition | Synonyms / aliases (by field) |
|---|---|---|
| Serial correlation | The tendency for measurements taken close together in time on the same person to be related rather than independent. | autocorrelation; serial dependence; within-subject / intra-subject correlation |
| Autocorrelation function (ACF) | A function or plot giving the correlation of a series with itself at each successive lag. | – |
| Lag | The number of time steps separating two observations being compared. | – |
| First-order autoregressive process, AR(1) | A model where each measurement correlates with the previous one and correlation decays as `rho` raised to the time gap. | AR(1); lag-1 autocorrelation; corAR1 / corCAR1 (continuous-time) |

| Term | Plain-language definition | Synonyms / aliases (by field) |
|---|---|---|
| ARMA / ARIMA | Time-series models combining autoregressive and moving-average (and, for ARIMA, differencing) terms. | autoregressive integrated moving average |
| Stationarity | The assumption that a series' mean, variance, and correlation structure stay constant over time. | stationary time series |
| Detrending | Removing a systematic upward/downward trend so fluctuations can be studied. | – |
| Pre-whitening | Removing autocorrelation from a series before analysis, which can accidentally remove a real effect. | whitening |
| Cochrane-Orcutt / Prais-Winsten | Older regression procedures that adjust estimates for first-order autocorrelated errors. | – |
| Effective sample size | The smaller number of independent observations that correlated data are statistically worth; shrinks under positive correlation. | effective number of independent observations |
| Slope-autocorrelation indeterminacy | The difficulty, in short series, of telling apart a genuine trend from trend produced by autocorrelated errors. | confounding of slope and autocorrelation |

## 6. Variance, power, and estimation

| Term | Plain-language definition | Synonyms / aliases (by field) |
|---|---|---|
| Components of variation | The separate sources of variability in an outcome (between-treatment, between-patient, patient-by-treatment, within-patient). | variance components; sources of variation |
| Between-patient variance | Variability in average outcomes from one patient to another. | between-subject variance; tau-squared (meta-analysis) |
| Within-patient variance | Variability across repeated occasions in the same patient on the same treatment. | within-subject / residual / error variance |
| Intraclass correlation coefficient (ICC) | The fraction of total variability due to differences between subjects; equivalently, how alike a subject's repeated measures are. | ICC; rho; intra-subject correlation |

| Term | Plain-language definition | Synonyms / aliases (by field) |
|---|---|---|
| Covariance / variance-covariance matrix | A table of the variances and pairwise correlations of repeated measures or parameter estimates. | Sigma / Omega matrix; dispersion matrix |
| Compound symmetry (CS) | A covariance pattern assuming every pair of repeated measures is equally correlated. | exchangeable correlation |
| Unstructured covariance | A covariance pattern estimating every variance and correlation freely with no imposed shape. | general covariance |
| Toeplitz covariance | A covariance pattern where the correlation depends only on how many timepoints apart two measures are. | banded covariance |
| Kronecker correlation | Building a large correlation matrix as the structured product of smaller ones (e.g., within-period x between-period). | Kronecker-product structure |
| Positive definite | A required property of a valid covariance matrix (no impossible negative variances). | PD |
| Heteroscedasticity | Unequal variance of the outcome across conditions or over time. | heteroskedasticity; unequal variances |
| Variogram | A plot of how the variance of differences between measurements grows with the time gap, used to detect autocorrelation. | semivariogram |
| Maximum likelihood estimation (MLE) | Choosing parameter values that make the observed data most probable. | ML; likelihood-based estimation |
| Restricted maximum likelihood (REML) | A maximum-likelihood variant that estimates variance components with less bias by adjusting for the fixed effects. | restricted / residual maximum likelihood |
| Likelihood ratio test (LRT) | A test comparing two nested models by the difference in their fit. | deviance chi-squared test |
| Wald test | A test judging whether a coefficient differs from zero using its estimate divided by its standard error. | Wald statistic |
| Kenward-Roger approximation | A small-sample degrees-of-freedom correction improving fixed-effect tests in mixed models. | KR correction |

| Term | Plain-language definition | Synonyms / aliases (by field) |
|---|---|---|
| Empirical Bayes / BLUP | A subject-specific estimate that borrows strength from the group mean, shrinking noisy individual estimates toward it. | best linear unbiased predictor; shrinkage estimate |
| Shrinkage | Pulling individual or extreme estimates toward a common value to reduce noise and overfitting. | regularization; penalization; regression to the mean; James-Stein shrinkage |
| Borrowing strength | Using data from other individuals or studies to sharpen a sparse unit's estimate. | partial pooling; shrinkage |
| Penalized splines / B-splines | Flexible curve-fitting building blocks with a roughness penalty, used to model smooth nonlinear effects without overfitting. | B-spline basis; smoothing / roughness penalty |
| Smoothing parameter (lambda) | A tuning knob controlling how wiggly versus smooth a fitted curve is. | penalization / tuning parameter |
| Non-centrality parameter | A quantity in a test statistic's distribution under the alternative hypothesis; larger values mean higher power. | NCP |
| Type I error | Falsely declaring an effect that does not exist (a false positive); its rate is the significance level alpha. | alpha; size of the test; false-positive rate |
| Type II error | Failing to detect a real effect (a false negative); its complement is power. | beta; false negative |
| Statistical power | The probability of correctly detecting a true effect of a given size. | 1 - beta; sensitivity of a test |
| Sample-size / power calculation | Computing how many patients (or periods/measurements) are needed for adequate power or precision. | power analysis; study planning |
| Effect size | A standardized measure of how large a treatment difference is, often in standard-deviation units. | standardized mean difference (SMD); Cohen's d; tau |
| Bias (of an estimate) | A systematic tendency for an estimate to be too high or too low on average. | systematic error |
| Mean squared error (MSE) | An accuracy summary combining an estimator's variance and squared bias. | RMSE (its square root); bias-variance trade-off |

| Term | Plain-language definition | Synonyms / aliases (by field) |
| --- | --- | --- |
| Confidence interval | A range that, over many repeats, would contain the true value a stated percentage of the time. | CI; interval estimate |
| Coverage | How often computed intervals actually contain the true value across repeated studies/simulations. | interval coverage |
| Monte Carlo simulation | Repeatedly generating random datasets with known truth to study how a method or design behaves. | simulation study |
| Multivariate normal distribution | A bell-shaped joint distribution for several correlated variables, set by a mean vector and covariance matrix, used to generate repeated-measures data. | MVN; joint normal |
| Bootstrap | Resampling the observed data (with replacement) to approximate the sampling distribution of a statistic. | parametric / wild / multiplier bootstrap |
| Pitman asymptotic relative efficiency | A ratio comparing how many subjects two designs need to reach the same power. | relative efficiency |
| Information criterion (AIC / BIC / QIC) | Scores ranking models by fit while penalizing extra parameters; lower is better (QIC is the GEE analogue of AIC). | Akaike / Bayesian / quasi-likelihood information criterion |

## 7. Bayesian methods

| Term | Plain-language definition | Synonyms / aliases (by field) |
| --- | --- | --- |
| Bayesian inference | Updating prior beliefs with data via Bayes' rule to obtain a posterior probability distribution over parameters. | Bayesian analysis |
| Prior distribution | The probability distribution encoding belief about a parameter before seeing the data. | prior |
| Posterior distribution | The updated distribution for a parameter after combining prior and data. | posterior |
| Posterior predictive distribution | The distribution of new/future data implied by the fitted model, used for prediction and model checking. | predictive distribution |

| Term | Plain-language definition | Synonyms / aliases (by field) |
|---|---|---|
| Noninformative prior | A deliberately vague prior that lets the data dominate. | flat / diffuse / weak / reference prior |
| Informative prior | A prior carrying real external knowledge or expert opinion. | – |
| Weakly informative prior | A prior that gently constrains estimates to plausible ranges without strongly steering results. | – |
| Conjugate prior | A prior whose form yields a posterior in the same family, simplifying computation. | (e.g., inverse-gamma prior for a variance) |
| Credible interval | A Bayesian interval containing the parameter with a stated probability given the data. | posterior interval; Bayesian confidence interval |
| Posterior probability of benefit | The Bayesian probability that a treatment's true effect is favourable, read off the posterior. | – |
| Hierarchical / partial pooling | Estimating group-specific values as a compromise between each group's own data and the overall average. | partial pooling; Bayesian shrinkage |
| Exchangeability | Treating individuals' effects as interchangeable draws from a common distribution. | – |
| Markov chain Monte Carlo (MCMC) | A simulation algorithm that draws samples from a complex posterior to enable Bayesian computation. | MCMC; Gibbs / Metropolis sampling; JAGS / WinBUGS / Stan estimation |
| Posterior predictive check | A model check comparing data simulated from the fitted model against the observed data. | Bayesian p-value |
| Prior elicitation | Formally turning experts' beliefs into a prior distribution, especially influential when data are scarce. | expert-opinion elicitation |
| Prior effective sample size | A way to express how much information a prior contributes by equating it to a number of observations. | prior ESS; ECSS / EHSS (effective current / historical sample size) |
| Prior-data conflict | Disagreement between the prior and the observed data, which can distort inference unless the prior discounts it. | prior-likelihood conflict; commensurability |
| Robust / mixture / power / commensurate prior | Priors designed to down-weight historical or external information when it conflicts with current data. | meta-analytic-predictive (MAP) prior; discounting prior |

| Term | Plain-language definition | Synonyms / aliases (by field) |
|---|---|---|
| Sequential probability ratio test (SPRT) | A method that tests a hypothesis after each new observation and stops once evidence is conclusive. | SPRT; bSPRT (bootstrapped); sequential analysis |

## 8. Missing data and dropout

| Term | Plain-language definition | Synonyms / aliases (by field) |
|---|---|---|
| Dropout / attrition | Participants leaving the study before completion, leaving incomplete data. | withdrawal; loss to follow-up; discontinuation; noncompletion |
| Differential dropout | Unequal (or systematically different) dropout between treatment arms. | differential / relative attrition |
| Biased / informative dropout | When those who leave differ systematically (e.g., responders quitting when switched to placebo), distorting results. | informative dropout |
| Missing completely at random (MCAR) | Missingness unrelated to any observed or unobserved data. | MCAR |
| Missing at random (MAR) | Missingness explainable by observed data, so it can be validly modelled / imputed. | ignorable missingness |
| Missing not at random (MNAR) | Missingness that depends on the unseen value itself and cannot be fully corrected from observed data. | nonignorable / informative missingness (IM) |
| Last observation carried forward (LOCF) | Filling a missing value with the patient's last recorded value; simple but often biased. | – |
| Multiple imputation (MI) | Filling missing values several times from their predicted distribution and combining results to reflect uncertainty. | imputation |
| Complete-case analysis | Analyzing only subjects with no missing data, which can bias results. | listwise deletion; completers analysis |
| Inverse probability weighting (IPW) | Reweighting observed cases to stand in for similar dropouts. | inverse-probability weights |
| Selection model | A missing-data model that predicts the chance of dropping out from the (possibly unseen) outcome. | – |
| Pattern-mixture model | A missing-data model letting completers and dropouts have different outcome distributions. | – |

| Term | Plain-language definition | Synonyms / aliases (by field) |
|---|---|---|
| Sensitivity analysis | Re-running an analysis under alternative assumptions to see how robust the conclusions are. | robustness analysis |

## 9. Single-case / psychometric effect measures

| Term | Plain-language definition | Synonyms / aliases (by field) |
|---|---|---|
| Level change | An abrupt shift in the average outcome from one phase to the next. | mean shift; phase jump; intercept shift |
| Trend / rate change | A change in the slope (rate of improvement or decline) between phases. | slope change; time-by-phase interaction; drift |
| Tau-U | A nonoverlap effect size for single-case data that quantifies improvement, optionally correcting for baseline trend. | – |
| Nonoverlap of all pairs (NAP) | The proportion of treatment-phase points exceeding baseline points across all pairwise comparisons. | – |
| Percentage of nonoverlapping data (PND) | The percentage of treatment points beyond the most extreme baseline point. | – |
| Percentage of all nonoverlapping data (PAND) | The proportion of data that would need removing to eliminate overlap between phases. | – |
| Standardized mean difference (SMD) | The mean difference between phases or treatments divided by a standard deviation, allowing comparison across scales. | between-case SMD; standardized effect size |
| Multilevel meta-analysis of single cases | Combining many single-case datasets in a hierarchical model to estimate average effects and their variation across cases. | MultiSCED-type analysis |
| Simulation Modelling Analysis (SMA) | A bootstrap-like resampling method generating null replicates to get empirical p-values for single-case data. | – |
| Piecewise / segmented regression | Fitting separate linear segments to different phases (baseline vs treatment) of a series. | spline / discontinuity growth model |
| Idiographic | Focused on patterns within a single individual. | within-person approach |

| Term | Plain-language definition | Synonyms / aliases (by field) |
|---|---|---|
| Nomothetic | Focused on general laws or averages across many individuals. | population approach |
| Ecological momentary assessment (EMA) | Repeatedly sampling a person's experiences in real time in their natural environment. | experience sampling |
| Reliability | The degree to which a measurement yields consistent, reproducible results. | (test-retest, internal-consistency reliability) |
| Standard error of measurement | The expected spread of measurement errors around a person's true score. | – |
| Measurement error | The discrepancy between an observed measurement and the true underlying value. | observation error; noise |
| Floor / ceiling effect | Scores clustering at the lowest or highest possible value, hiding true differences. | – |

## 10. Biomarkers, precision medicine, and treatment-effect heterogeneity

| Term | Plain-language definition | Synonyms / aliases (by field) |
|---|---|---|
| Precision medicine | Tailoring treatment to an individual's biological, genetic, physiological, and behavioural profile. | personalized / personalised / individualized / stratified / tailored medicine |
| Patient-by-treatment interaction | Genuine variation among patients in how strongly they respond to a treatment; the core target of personalized medicine. | treatment-effect heterogeneity (HTE); subject-by-treatment interaction; differential / individual response; effect modification |
| Heterogeneity of treatment effects (HTE) | The fact that the same treatment helps different patients by different amounts. | treatment-effect heterogeneity; differential treatment effect |
| Effect modifier | A patient characteristic that changes the size or direction of a treatment's effect. | moderator (psychology); interaction variable; predictive marker |
| Biomarker-treatment interaction | The statistical effect showing a treatment's benefit depends on a patient's biomarker value; the key signal these designs test. | treatment-by-biomarker interaction; predictive interaction; effect modification |

| Term | Plain-language definition | Synonyms / aliases (by field) |
|---|---|---|
| Predictive biomarker | A measurable patient trait that forecasts who will benefit (differentially) from a specific treatment and guides its selection. | predictive marker / signature / covariate; treatment-effect modifier; theranostic biomarker |
| Prognostic biomarker | A trait that predicts the likely course of disease regardless of treatment. | prognostic factor / variable / covariate; baseline covariate |
| Qualitative (biologic) interaction | An interaction where treatment helps some patients but harms others (the effect reverses direction); cannot be removed by rescaling. | crossover / non-removable interaction |
| Quantitative (statistical) interaction | An interaction in degree only (benefit varies in size but keeps direction); often removable by transforming the scale. | removable / scale-dependent interaction |
| Additivity | When the treatment effect is constant across patients on a chosen scale, so there is no interaction. | additive model |
| Individual treatment effect (ITE) | The difference in outcome a specific patient would have under treatment versus control. | individual-level effect |
| Conditional average treatment effect (CATE) | The average treatment effect within a subgroup defined by particular characteristics. | – |
| Average treatment effect (ATE) | The mean treatment effect across the whole population. | population / main effect |
| Subgroup analysis | Examining treatment effects within predefined patient subsets; prone to spurious findings if unplanned. | subgroup comparison |
| Risk modeling (predictive HTE) | Stratifying patients by their predicted baseline risk to see who benefits most. | risk-stratified / risk-based subgrouping |
| Effect modeling (predictive HTE) | Building a model with treatment-by-covariate interactions to predict individual benefit. | interaction-based modeling |
| Responder / non-responder | A patient classified as benefiting or not, often by a threshold. | response classification |
| Biomarker-positive / -negative subgroup (B+ / B-) | The patients whose biomarker indicates they are more (B+) or less (B-) likely to benefit. | targeted / non-targeted subgroup |
| Biomarker prevalence | The fraction of the population that is biomarker-positive. | biomarker frequency |

| Term | Plain-language definition | Synonyms / aliases (by field) |
|---|---|---|
| Companion diagnostic | A laboratory test required to identify patients eligible for a specific targeted therapy. | – |
| Potential outcomes / counterfactual | The pair of outcomes a unit would show under each treatment (only one is observed); the counterfactual is the unobserved one. | Neyman-Rubin model; potential outcome |
| SUTVA | The assumption that one unit's treatment does not affect another's outcome and there is only one version of each treatment. | stable unit treatment value assumption |
| Pharmacogenomics | Using a patient's genetics to predict and tailor treatment response. | individualized / precision therapy |
| Learning health system | An infrastructure that continually uses routine clinical data to learn which patients benefit from which treatments. | rapid-learning health system |

## 11. Clinical, pharmacology, and outcome terms

| Term | Plain-language definition | Synonyms / aliases (by field) |
|---|---|---|
| Pharmacokinetic / pharmacodynamic (PK/PD) model | Models of how a drug's concentration changes over time (PK) and how concentration produces an effect (PD). | PK/PD model; Emax / dose-response model |
| Half-life | The time for a drug's concentration (or effect) to fall by half; governs carryover length. | t-half; elimination rate; clearance |
| Cumulative dose effect | A drug effect that depends on the whole dosing history, weighted toward recent doses. | dose-history dependence |
| Modified Gompertz function | A three-parameter S-shaped curve (maximum response, rate, displacement) modelling a response rising to a plateau. | three-parameter Gompertz |
| Exponential decay | A model where an effect shrinks by half over a fixed time, used for carryover persistence (`exp(-lambda * t)`). | half-life / lambda decay |

| Term | Plain-language definition | Synonyms / aliases (by field) |
| --- | --- | --- |
| Regression to the mean | The tendency of extreme initial values to drift toward average on re-measurement, mimicking improvement. | – |
| Bioequivalence | Showing two drug formulations deliver the active ingredient similarly enough to be interchangeable. | average / population / individual bioequivalence |
| Prescribability / switchability | Whether a generic can be given to new patients (prescribability) or substituted in patients already on the brand (switchability). | population vs individual equivalence |
| Minimal clinically important difference (MCID) | The smallest change in an outcome that patients perceive as meaningful. | clinical-importance cut-off |
| Clinical vs statistical significance | Whether an effect is large enough to matter to a patient versus merely unlikely to be chance. | clinical significance / importance |
| Patient-reported outcome measure (PROM) | An outcome reported directly by the patient (symptoms, quality of life). | PRO; self-report outcome |
| Surrogate outcome | An early or proxy measurement standing in for the true outcome of interest. | surrogate endpoint |
| Time-to-event endpoint | An outcome measured as the time until an event occurs, analyzed with survival methods. | survival endpoint; PFS / OS |
| Hazard rate / hazard ratio | The instantaneous risk of an event over time; the ratio comparing that risk between groups. | HR; baseline hazard |
| Absolute risk difference | The plain difference in outcome probability between treated and control patients. | risk difference; absolute treatment effect |
| Relative risk / risk ratio | The ratio of event probability in one group to another. | RR; relative risk reduction (its complement) |
| Odds ratio | A measure comparing the odds of an outcome between two groups. | OR |
| Censoring | When the event of interest is not observed within follow-up (e.g., the study ends first). | administrative / right censoring |
| Comparative effectiveness research (CER) | Research comparing real-world treatments head-to-head to see which works better for whom. | head-to-head comparison; patient-centered outcomes research (PCOR) |

| Term | Plain-language definition | Synonyms / aliases (by field) |
|---|---|---|
| Evidence-based medicine (EBM) | Practicing medicine by applying the best current research evidence to patient care. | – |
| Therapeutic window | The dose range that is effective without causing harmful side effects. | therapeutic range |
| Rare / ultra-rare disease | A condition affecting so few people that conventional group trials are infeasible, favouring N-of-1 designs. | low-volume treatment context |
| Actigraphy | Using a wrist-worn motion sensor to objectively estimate sleep and activity. | actigraph; accelerometry |
| Prazosin | An alpha-1 adrenoceptor antagonist studied for PTSD trauma nightmares and sleep disturbance (the project's running clinical example). | alpha-1 blocker |
| PTSD / CAPS | Post-traumatic stress disorder; the Clinician-Administered PTSD Scale (CAPS) is a structured interview rating its symptom frequency and intensity. | post-traumatic stress disorder; CAPS-IV |

## 12. Cross-field synonym map (same concept, different name)

The single concept on the left is named differently depending on whether a paper is written from a biostatistics, single-case / psychometric, or precision-medicine perspective.

| Underlying concept | Biostatistics term | Single-case / psychometric term | Precision-medicine term |
|---|---|---|---|
| One-patient repeated-crossover study | N-of-1 trial; single-patient trial | single-case experimental design (SCED); single-subject design | idiographic / individualized clinical trial |
| Genuine person-to-person variation in response | patient-by-treatment interaction; variance component | subject-by-treatment interaction | heterogeneity of treatment effects (HTE) |
| A variable that changes a treatment's effect | treatment-by-covariate interaction; effect modification | moderator | predictive biomarker / effect modifier |
| Subject-specific deviations in a model | random effects; variance components | between-case variance | (individual effects) |

| Underlying concept | Biostatistics term | Single-case / psychometric term | Precision-medicine term |
|---|---|---|---|
| Pulling individual estimates toward the mean | shrinkage; empirical Bayes / BLUP | – | borrowing strength / partial pooling |
| Correlation of nearby-in-time measures | serial correlation; AR(1); ICC | autocorrelation | – |
| Combining many small studies/patients | mixed-model / IPD meta-analysis | multilevel meta-analysis of single cases | aggregated N-of-1 analysis |
| The unobserved alternative outcome | counterfactual; potential outcome | – | individual treatment effect (ITE) |
| Multiple measurements per unit over time | repeated measures; longitudinal data | single-case time series | – |
| Standardized magnitude of an effect | standardized mean difference; Cohen's d | Tau-U / NAP / PND (nonoverlap indices) | – |
| Real-time repeated outcome capture | – | ecological momentary assessment (EMA) | micro-randomized trial sampling |

## 13. Reporting guidelines and ethics

| Term | Plain-language definition | Synonyms / aliases (by field) |
|---|---|---|
| CONSORT | The standard checklist for transparently reporting randomized controlled trials. | Consolidated Standards of Reporting Trials |
| CENT 2015 | The CONSORT extension giving reporting standards specifically for N-of-1 trials. | CONSORT Extension for N-of-1 Trials |
| SPENT / SPIRIT | SPIRIT is the trial-protocol reporting guideline; SPENT is its N-of-1 protocol extension. | SPIRIT extension for N-of-1 |
| RoBiNT scale | A risk-of-bias / methodological-quality rating scale for single-case and N-of-1 studies. | Risk of Bias in N-of-1 Trials scale |
| PATH statement | The Predictive Approaches to Treatment effect Heterogeneity reporting guidance for HTE analyses. | – |
| TRIPOD / PROGRESS / ICEMAN | Reporting/appraisal frameworks for prediction models (TRIPOD), prognosis research (PROGRESS), and credibility of subgroup/interaction claims (ICEMAN). | – |

| Term | Plain-language definition | Synonyms / aliases (by field) |
|---|---|---|
| EQUATOR Network | An organization promoting accurate, transparent reporting of health research via guidelines. | Enhancing Quality and Transparency Of health Research |
| Operating characteristics | A design's statistical performance profile (type I error, power, bias) across scenarios. | frequentist properties |
| Pre-specification (a priori analysis) | Defining hypotheses and analyses before seeing data, to avoid data-driven false findings. | pre-registration |
| Selective reporting / data dredging | Cherry-picking favourable analyses or subgroups, inflating false positives. | multiplicity; subgroup-fishing |
| Family-wise error rate (FWER) | The probability of at least one false positive across multiple tests. | multiplicity control |
| Alpha allocation / spending | Dividing the overall significance level across hypotheses or interim looks to control multiplicity. | alpha splitting / spending; gatekeeping; fixed-sequence testing |
| Internal vs external validity | Whether conclusions are valid within the studied sample (internal) versus generalize beyond it (external). | generalizability; applicability; representativeness |
| Equipoise | Genuine uncertainty about which treatment is better, ethically justifying randomization. | clinical equipoise |
| Informed consent / IRB approval | Ethical safeguards requiring participant agreement and institutional ethics-board review before research. | ethics-committee approval; written consent |
| Patient and public involvement (PPI) | Involving patients and the public in designing, running, and interpreting research. | patient representatives |

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## Provenance and epistemic status

- Corpus. The `nof1` collection of the project Zotero library holds 72 curated N-of-1 / precision-medicine articles. Local full text was obtained for 46; 42 were successfully machine-read and mined for terms (4 were skipped owing to a content-filter false positive on one processing batch, but their concepts are redundantly covered by other articles). Three of the read files held mismatched or banner-only text (an unrelated BMJ news page, an arbovirus review, and a publisher download banner) and contributed no terms; one file was mislabelled in the catalogue (a Bayesian effective-sample-size paper) and was mined on its actual content.
- Definitions were extracted by reading the source texts; plain-language wording is the lexicon authors' paraphrase, verified against the mined passages. The statistical / psychometric supplement (Sections 5-9 in part)

draws on standard methodology references (Pinheiro & Bates; Fitzmaurice, Laird & Ware; Box, Jenkins & Reinsel; Gelman et al.; Imbens & Rubin; de Vet et al.; CENT; Senn) and is inferred from established domain knowledge rather than verified against a specific retrieved source.

- Synonym groupings reflect usage observed across the corpus plus conventional cross-field equivalences; they are aids to recognition, not claims of exact mathematical identity (e.g., a quantitative interaction and an additive-scale effect coincide only on a particular scale).

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